

**MEDIATEK**

# MT8676 Android BT User Manual

**Version:** 1.0  
**Release date:** 2024-08-12

Use of this document and any information contained therein is subject to the terms and conditions set forth in Exhibit 1. This document is subject to change without notice.

---

## Version History

| Version | Date       | Author     | Description      |
|---------|------------|------------|------------------|
| 1.0     | 2023-08-12 | Liuqin Liu | Official release |

Table of Contents

Version History ..... 2

Table of Contents ..... 3

List of Figures ..... 3

List of Tables ..... 3

1 BT ..... 4

1.1 Overview ..... 4

1.1.1 Brief Introduction ..... 4

1.1.2 BT Abbreviations ..... 4

1.2 Architecture/Process Overview ..... 4

1.3 Configuration/Customization Guideline ..... 5

1.3.1 Profile Configuration ..... 5

1.4 Frequently Asked Questions/Troubleshooting ..... 6

1.4.1 Supported Bluetooth Feature Set And Certification ..... 6

1.4.2 BQB ..... 6

1.4.3 BT Issue Submission SOP ..... 7

1.4.4 BT Music (A2DP) Issue ..... 8

1.4.5 BT Scan Issue ..... 9

1.4.6 BT Connect Issue ..... 10

1.4.7 BT Voice Issue ..... 10

Exhibit 1 Terms and Conditions ..... 12

List of Figures

Figure 1-1. BT architecture ..... 5

Figure 1-2. BT profile configuration ..... 6

Figure 1-3. BT issue submission SOP ..... 8

List of Tables

Table 1-1. Abbreviations ..... 4

# 1 BT

## 1.1 Overview

### 1.1.1 Brief Introduction

This section introduces the basic functions of Bluetooth (BT) and the debugging methods of common Bluetooth problems. The BT chip of the MT8676 platform is MT6637.

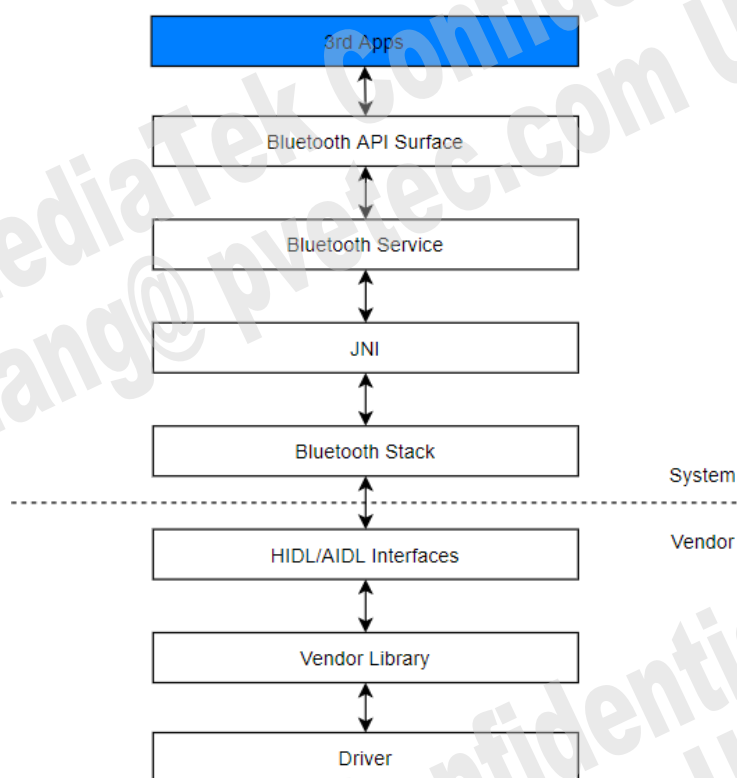
### 1.1.2 BT Abbreviations

Table 1-1. Abbreviations

| Abbreviation | Explanation                         |
|--------------|-------------------------------------|
| A2DP         | Advanced Audio Distribution Profile |
| AVRCP        | Audio/Video Remote Control Profile  |
| BQB          | Bluetooth Qualification Body        |
| BT           | Bluetooth                           |
| DUT          | Device Under Test                   |
| FW           | Firmware                            |
| HCI          | Host Controller Interface           |
| HFP          | Hands-free Profile                  |
| LE           | Low Energy                          |

## 1.2 Architecture/Process Overview

The MT8676 BT architecture is shown below:



**Figure 1-1. BT architecture**

## 1.3 Configuration/Customization Guideline

### 1.3.1 Profile Configuration

BT has different application scenarios, such as a2dp sink/hfp client/avrcp ct profiles. Android U version supports configuring different profiles to meet various application scenarios. You can configure it by adding

“PRODUCT\_PROPERTY\_OVERRIDES” in device-vext.mk. The path to MediaTek’s public version file is:

device/mediateksample/<project name>/ device-vext.mk

device/mediatek/<chip name>/ device-vext.mk

```

40 # Bluetooth IVI-Driver AOSP Property
41 PRODUCT_PROPERTY_OVERRIDES += bluetooth.profile.a2dp.sink.enabled=true
42 PRODUCT_PROPERTY_OVERRIDES += bluetooth.profile.a2dp.source.enabled=false
43 PRODUCT_PROPERTY_OVERRIDES += bluetooth.profile.asha.central.enabled=false
44 PRODUCT_PROPERTY_OVERRIDES += bluetooth.profile.avrcp.controller.enabled=true
45 PRODUCT_PROPERTY_OVERRIDES += bluetooth.profile.avrcp.target.enabled=false
46 PRODUCT_PROPERTY_OVERRIDES += bluetooth.profile.bap.broadcast.assist.enabled=false
47 PRODUCT_PROPERTY_OVERRIDES += bluetooth.profile.bap.broadcast.source.enabled=false
48 PRODUCT_PROPERTY_OVERRIDES += bluetooth.profile.bap.unicast.client.enabled=false
49 PRODUCT_PROPERTY_OVERRIDES += bluetooth.profile.bas.client.enabled=false
50 PRODUCT_PROPERTY_OVERRIDES += bluetooth.profile.bass.client.enabled=false
51 PRODUCT_PROPERTY_OVERRIDES += bluetooth.profile.csip.set_coordinator.enabled=false
52 PRODUCT_PROPERTY_OVERRIDES += bluetooth.profile.gatt.enabled=true
53 PRODUCT_PROPERTY_OVERRIDES += bluetooth.profile.hap.client.enabled=false
54 PRODUCT_PROPERTY_OVERRIDES += bluetooth.profile.hfp.ag.enabled=false
55 PRODUCT_PROPERTY_OVERRIDES += bluetooth.profile.hfp.hf.enabled=true
56 PRODUCT_PROPERTY_OVERRIDES += bluetooth.profile.hid.device.enabled=false
57 PRODUCT_PROPERTY_OVERRIDES += bluetooth.profile.hid.host.enabled=true
58 PRODUCT_PROPERTY_OVERRIDES += bluetooth.profile.map.client.enabled=true
59 PRODUCT_PROPERTY_OVERRIDES += bluetooth.profile.map.server.enabled=false
60 PRODUCT_PROPERTY_OVERRIDES += bluetooth.profile.mcp.server.enabled=false
61 PRODUCT_PROPERTY_OVERRIDES += bluetooth.profile.opp.enabled=true
62 PRODUCT_PROPERTY_OVERRIDES += bluetooth.profile.pan.nap.enabled=false
63 PRODUCT_PROPERTY_OVERRIDES += bluetooth.profile.pan.panu.enabled=false
64 PRODUCT_PROPERTY_OVERRIDES += bluetooth.profile.pbap.client.enabled=true
65 PRODUCT_PROPERTY_OVERRIDES += bluetooth.profile.pbap.server.enabled=false
66 #PRODUCT_PROPERTY_OVERRIDES += bluetooth.profile.sap.server.enabled=true
67 PRODUCT_PROPERTY_OVERRIDES += bluetooth.profile.ccp.server.enabled=false
68 PRODUCT_PROPERTY_OVERRIDES += bluetooth.profile.vcp.controller.enabled=false
69 PRODUCT_PROPERTY_OVERRIDES += bluetooth.profile.map_client.sms_receiver_package=com.mediatek.bluetooth

```

**Figure 1-2. BT profile configuration**

## 1.4 Frequently Asked Questions/Troubleshooting

### 1.4.1 Supported Bluetooth Feature Set And Certification

The controller and MediaTek stack of the MT8676 + MT6637 platform can pass the Bluetooth 5.4 certification.

Most automotive customers use a third-party stack and need to confirm the certification version with the third-party stack.

#### 1.4.2 BQB

##### 1.4.2.1 What is BQB

BQB: The full name is Bluetooth Qualification Body, which is generally referred to as Bluetooth Certification.

### 1.4.2.2 Why to Pass BQB

Bluetooth certification is a procedure that any product using Bluetooth wireless technology must go through. Only after being tested and certified by the Special Interest Group (SIG) as complying with Bluetooth standards, is it eligible to be put on the market under the name of a Bluetooth product. Otherwise, it is illegal.

### 1.4.2.3 Basic Introduction to BQB Certification Process

Automotive products generally use 3rd party BT Stack. During certification, only the certification of BT Controller requires MediaTek's assistance. BT Stack layer certification, such as Profile certification, is the responsibility of the 3rd party (if MediaTek stack is used, MediaTek is responsible). If the certification center is not familiar with MediaTek's solution, it may ask for certification test methods and log capture methods. You can ask IDH for assistance, because IDH's own core board has BT test projects that can be used as a reference.

### 1.4.2.4 Certification Issue Handling Process

1. The certification center reports the problem
2. Tier 1 sorts out the problems and distinguishes whether it is MediaTek issue or 3rd party issue
3. MediaTek -related issue are submitted to IDH for reproduction, and IDH's own laboratory reproduces the problem first
4. If IDH needs MediaTek's assistance, please submit CR.

### 1.4.3 BT Issue Submission SOP

If you are using a third-party stack, you need first-hand analysis from the third-party stack. If the analysis shows that it is an MediaTek's problem, please create a CR and provide the corresponding log (please consult the third-party stack for the capture method) and inform the time of the problem, and attach the analysis process of the third-party stack.

Please refer to [Figure 1-3](#) for BT issue submission SOP.

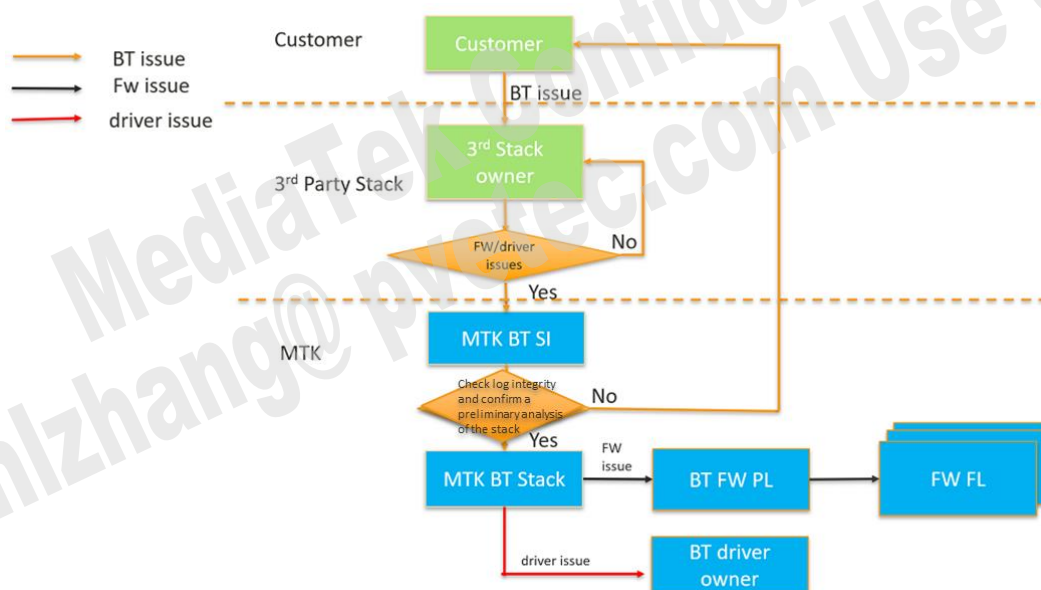


Figure 1-3. BT issue submission SOP

#### 1.4.4 BT Music (A2DP) Issue

##### 1.4.4.1 Symptom

- A2DP sink/source stuttering
- A2DP no sound

##### 1.4.4.2 Check Point [Stack Check]

- A2DP Source: The interval between each two TX A2DP data packets' NOCP (Number of Complete Packet event) exceeds 80ms.  
Case 1: After TX data, NOCP comes back late? Yes -> MediaTek check  
Case 2: After the previous NOCP packet comes back, the next stack TX data packet is late? Yes -> Stack check
- A2DP Sink: The interval between each two RX A2DP data packets is greater than 80ms. Confirm the peer sending interval through Air log, if OK -> MediaTek check

##### 1.4.4.3 Note

- In other cases, the host and FW need to confirm and explain the reasons, and both parties must reach a consensus.
- Please provide hci/fw/air log.

## 1.4.5 BT Scan Issue

### 1.4.5.1 Symptom

- Unable to search for the other device (Inquiry fail/LE scan fail)
- Cannot be searched by the other device (Inquiry scan/LE adv abnormal)

### 1.4.5.2 Check Point [Stack Check]

- No device found:  
Is there an enable inquiry command in the HCI log?  
Yes -> MediaTek check
- No device scanned via LE:  
Is there a command to enable BLE scan in the HCI log during the scan failure period?  
Yes -> MediaTek check
- Inquiry to a device takes too long (performance):  
Describe the start and end time of the inquiry, and whether the host sent an inquiry command during this period?  
Yes -> MediaTek check
- The peers cannot find DUT:  
During the search period, whether there is a command to enable inquiry scan (BT search) or enable advertising (LE search) in HCI log?  
Yes -> MediaTek check

### 1.4.5.3 Note

- Understand the notes information:
  - (1) Are there other devices that can be searched (Inquiry fail/LE scan fail)
  - (2) If the DUT cannot find the device, can it be found using other devices (Inquiry scan/LE adv)
  - (3) The maximum number of responses that can be received before the end, and whether the total inquiry time is normal (no device can be inquired)
  - (4) Whether the event is reported, when the DUT cannot search for the device, whether it can be searched with other devices (no device can be inquired)
- Log requirements:
  - (1) BT FW log+ hci log + air log
  - (2) Provide the exact time when the problem occurred

## 1.4.6 BT Connect Issue

### 1.4.6.1 Symptom

- ACL/LE link establishment failed (Page timeout/page scan/LE init/LE adv)

### 1.4.6.2 Check Point [Stack check]

- Connection Failed
  - During the connection establishment period, is the peer reconnecting (near the page timeout, is there a connection request event caused by the peer), if so, it is considered normal
  - Page timeout (how long will the page timeout be in a log? After confirming that the exception is caused by page timeout, MediaTek check)
  - Occasionally, the HCI\_Disconnection\_Complete Reason is Connection Failed to be Established (0x3E). This is usually a normal phenomenon. It is more likely to occur in scenarios where the broadcast channel (37/38/39) is congested or interfered. Please first confirm whether this case is still likely to occur in a clean environment. If so, MediaTek check.

### 1.4.6.3 Note

- Connect issues are determined by whether the connection is established in the HCI log (connection complete event (status: success)). If the profile connection fails after the connection is established, the Stack needs to first identify the specific cause of the failure. If it is confirmed to be related to FW (Data no response? Data Transfer problem...), then provide relevant analysis and logs and ask MediaTek to check.
- MediaTek check log requirements:  
BT FW log+ hci log + air log  
Provide the exact time when the problem occurred

## 1.4.7 BT Voice Issue

### 1.4.7.1 Symptom

- SCO/eSCO link not connected, noise, or no sound, etc.

### 1.4.7.2 Check Point

- HFP connection failed/disconnection failed [Stack check]

The command from the host does not get the correct event or the profile does not get the correct response. Check the settings of the SCO/eSCO link (including whether the codec/packet type and profile settings are consistent).

- HFP sound is silent/stuttering/noise [Audio check]  
Download (DL): Host (Audio) -> BT FW -> Air -> peer device -> remote auxiliary machine  
Upload (UL): remote auxiliary machine -> peer device -> Air -> BT FW -> Host (Audio)

#### 1.4.7.3 Note

- If the sound you hear during a call with the remote phone is just background noise (air log), the original HFP call sound is coherent and there is no interruption, and the BT FW serves the anchor point normally => audio problem, audio check.

## Exhibit 1 Terms and Conditions

Your access to and use of this document and the information contained herein (collectively this “Document”) is subject to your (including the corporation or other legal entity you represent, collectively “You”) acceptance of the terms and conditions set forth below (“T&C”). By using, accessing or downloading this Document, You are accepting the T&C and agree to be bound by the T&C. If You don’t agree to the T&C, You may not use this Document and shall immediately destroy any copy thereof.

This Document contains information that is confidential and proprietary to MediaTek Inc. and/or its affiliates (collectively “MediaTek”) or its licensors and is provided solely for Your internal use with MediaTek’s chipset(s) described in this Document and shall not be used for any other purposes (including but not limited to identifying or providing evidence to support any potential patent infringement claim against MediaTek or any of MediaTek’s suppliers and/or direct or indirect customers). Unauthorized use or disclosure of the information contained herein is prohibited. You agree to indemnify MediaTek for any loss or damages suffered by MediaTek for Your unauthorized use or disclosure of this Document, in whole or in part.

MediaTek and its licensors retain titles and all ownership rights in and to this Document and no license (express or implied, by estoppels or otherwise) to any intellectual propriety rights is granted hereunder. This Document is subject to change without further notification. MediaTek does not assume any responsibility arising out of or in connection with any use of, or reliance on, this Document, and specifically disclaims any and all liability, including, without limitation, consequential or incidental damages.

THIS DOCUMENT AND ANY OTHER MATERIALS OR TECHNICAL SUPPORT PROVIDED BY MEDIATEK IN CONNECTION WITH THIS DOCUMENT, IF ANY, ARE PROVIDED “AS IS” WITHOUT WARRANTY OF ANY KIND, WHETHER EXPRESS, IMPLIED, STATUTORY, OR OTHERWISE. MEDIATEK SPECIFICALLY DISCLAIMS ALL WARRANTIES OF MERCHANTABILITY, NON-INFRINGEMENT, FITNESS FOR A PARTICULAR PURPOSE, COMPLETENESS OR ACCURACY AND ALL WARRANTIES ARISING OUT OF TRADE USAGE OR OUT OF A COURSE OF DEALING OR COURSE OF PERFORMANCE. MEDIATEK SHALL NOT BE RESPONSIBLE FOR ANY MEDIATEK DELIVERABLES MADE TO MEET YOUR SPECIFICATIONS OR TO CONFORM TO A PARTICULAR STANDARD OR OPEN FORUM.

Without limiting the generality of the foregoing, MediaTek makes no warranty, representation or guarantee regarding the suitability of its products for any particular purpose, nor does MediaTek assume any liability arising out of the application or use of any product, circuit or software. You agree that You are solely responsible for the designing, validating and testing Your product incorporating MediaTek’s product and ensure such product meets applicable standards and any safety, security or other requirements.

The above T&C and all acts in connection with the T&C or this Document shall be governed, construed and interpreted in accordance with the laws of Taiwan, without giving effect to the principles of conflicts of law.